

Lecture Note 12: Bayes Procedures

14.380: Statistical Methods in Economics, Fall 2026

1 Frequentist and Bayesian Paradigms

According to frequentist theory, it is assumed that the unknown parameter θ is some fixed number or vector. Given parameter value θ , we observe sample data X from distribution $f(\cdot|\theta)$. To estimate parameter θ , we introduce an estimator $T(X)$ which is a statistic, i.e. function of the data. This statistic is a random variable, since X is random. That is, the randomness here comes from randomness of sampling. Different properties of estimation characterize this randomness. For example, the concept of unbiasedness means that if we observe many samples from distribution $f(\cdot|\theta)$, then the average of $T(X)$ over the samples will be close to the true parameter value θ , i.e. $\mathbb{E}_\theta[T(X)] = \theta$. The concept of consistency means that if we observe many samples from distribution $f(\cdot|\theta)$, then the distribution of $T(X)$ over these samples will be close in probability to the true parameter value θ , at least when we observe large samples, i.e. $T(X)$ will be in a small neighborhood of θ in most observed samples.

In contrast, Bayesian theory assumes that θ is some random variable and we are interested in the realization of this random variable. This realization, say, θ_0 , is thought to be the true parameter value. It is assumed that we know the distribution of θ , or at least its approximation. This distribution is called a prior. It usually comes from our subjective belief based on our past experience. Once θ_0 is realized, we observe a random sample $X = (X_1, \dots, X_n)$ from distribution $f(\cdot|\theta_0)$. Once we have data, the best thing we can do is to calculate the conditional distribution of θ given $X_1, \dots, X_n$. This conditional distribution is called the *posterior*. The posterior is used to create an estimate for θ . Since we condition on the observations X_i , we treat them as given. The randomness in the posterior is uncertainty about θ . The Bayesian approach is often used in learning theory.

2 Bayesian Updating

Let $\pi(\theta)$ denote our prior. In other words, the parameter of interest θ is a random variable with distribution $\pi(\cdot)$. Our model is $f(\cdot|\theta)$. In other words, once we have a realization of a parameter value θ , the observed sample data X are drawn from the conditional distribution $f(\cdot|\theta)$. Then the joint pdf of θ and X is $f(x, \theta) = \pi(\theta)f(x|\theta)$. The prior predictive distribution is $m(x) = \int \pi(\tilde{\theta})f(x|\tilde{\theta})d\tilde{\theta}$. Thus, the prior predictive distribution is the marginal pdf of our sample data X . Once we observe $X = x$, we can calculate the posterior distribution for θ as

$$\pi(\theta|X = x) = \frac{f(x, \theta)}{m(x)} = \frac{\pi(\theta)f(x|\theta)}{\int \pi(\tilde{\theta})f(x|\tilde{\theta})d\tilde{\theta}}.$$

Example Let $X = (X_1, \dots, X_n)$ be a random sample from a Bernoulli(p) distribution. Then the joint pdf of the data is $f(x|p) = p^{\sum_i x_i}(1-p)^{n-\sum_i x_i}$ where $x = (x_1, \dots, x_n)$. In classical

theory, we would consider some estimator of p . For example, we can take $T(X) = \bar{X}_n$. Then we have already seen that $\mathbb{E}_p[T(X)] = p$ and $T(X) \rightarrow_p p$. In Bayesian theory we need some prior distribution of p . For example, suppose we believe that all p are equally possible. Then we have a uniform prior, i.e. $\pi(p) = 1$ if $p \in [0, 1]$ and 0 otherwise. Then the joint pdf of p and X is $f(x, p) = p^{\sum_i x_i} (1 - p)^{n - \sum_i x_i} I\{0 \leq p \leq 1\}$. The prior predictive distribution is

$$m(x) = \int_0^1 p^{\sum_i x_i} (1 - p)^{n - \sum_i x_i} dp = B\left(\sum_{i=1}^n x_i + 1, n - \sum_{i=1}^n x_i + 1\right),$$

where $B(x, y) = \int_0^1 t^{x-1} (1 - t)^{y-1} dt$ is the Beta-function. The posterior distribution is

$$\pi(p|X = x) = \frac{p^{\sum_i x_i} (1 - p)^{n - \sum_i x_i}}{B\left(\sum_{i=1}^n x_i + 1, n - \sum_{i=1}^n x_i + 1\right)} I\{0 \leq p \leq 1\}.$$

This distribution is called the *Beta* $\mathcal{B}(\alpha, \beta)$ distribution with parameters $\alpha = \sum_{i=1}^n x_i + 1$ and $\beta = n - \sum_{i=1}^n x_i + 1$. Its mean is

$$\mathbb{E}[p|X = x] = \frac{\alpha}{(\alpha + \beta)} = \frac{\sum_{i=1}^n x_i + 1}{n + 2}.$$

Its variance is

$$V(p|X = x) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)} = \frac{(\sum_{i=1}^n x_i + 1)(n - \sum_{i=1}^n x_i + 1)}{(n + 2)^2(n + 3)}.$$

2.1 How to calculate posterior distribution

Here we consider a trick that may help to calculate the posterior analytically. It does not always work, though.

Let $X = (X_1, \dots, X_n)$ be a random sample from an $N(\mu, \sigma^2)$ distribution. Suppose σ^2 is known. Let $N(\mu_0, \tau^2)$ be the prior distribution for μ . Then the posterior distribution is $\pi(\mu|X = x) = \pi(\mu)f(x|\mu)/m(x)$ where $\pi(\mu)$ denotes the prior distribution and $m(x)$ denotes the prior predictive distribution. Note that $m(x)$ does not depend on μ . So $m(x)$ is just a constant that normalizes $\int \pi(\mu|X = x)d\mu$ to 1. Therefore, when we calculate $\pi(\mu)f(x|\mu)$, we can denote all multiplicative terms which do not contain μ as some constant C instead of keeping track of all these terms. Once we have an expression for $\pi(\mu)f(x|\mu)$ as a function of μ , we can integrate it in order to find the normalizing constant $m(x)$. In our case,

$$\pi(\mu|X = x) = C\pi(\mu)f(x|\mu) = C \exp \left\{ -\frac{\sum_{i=1}^n (x_i - \mu)^2}{2\sigma^2} - \frac{(\mu - \mu_0)^2}{2\tau^2} \right\},$$

where C contains all terms which do not contain μ . Thus,

$$\begin{aligned}
\pi(\mu|X=x) &\propto \exp\left\{\frac{\mu\sum_{i=1}^n x_i}{\sigma^2} - \frac{n\mu^2}{2\sigma^2} - \frac{\mu^2}{2\tau^2} + \frac{\mu\mu_0}{\tau^2}\right\} \\
&\propto \exp\left\{-\mu^2\left(\frac{n}{2\sigma^2} + \frac{1}{2\tau^2}\right) + 2\mu\left(\frac{\sum_{i=1}^n x_i}{2\sigma^2} + \frac{\mu_0}{2\tau^2}\right)\right\} \\
&\propto \exp\left\{-\frac{1}{2}\left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)\left[\mu^2 - 2\mu\left(\frac{\sum_{i=1}^n x_i/\sigma^2 + \mu_0/\tau^2}{n/\sigma^2 + 1/\tau^2}\right)\right]\right\} \\
&\propto \exp\left\{-\frac{1}{2}\left(\frac{n}{\sigma^2} + \frac{1}{\tau^2}\right)[\mu - \tilde{\mu}]^2\right\} \\
&\propto \exp\left\{-\frac{(\mu - \tilde{\mu})^2}{2\tilde{\sigma}^2}\right\},
\end{aligned}$$

where $\tilde{\mu} = (\sum_{i=1}^n x_i/\sigma^2 + \mu_0/\tau^2)/(n/\sigma^2 + 1/\tau^2)$ and $\tilde{\sigma}^2 = 1/(n/\sigma^2 + 1/\tau^2)$. Note that, via some abuse of notation, symbol $\propto$ stands for “proportional as a function of μ ”, thus different lines need different normalizing constants. Thus, the conditional distribution of μ given $X = x$ is $N(\tilde{\mu}, \tilde{\sigma}^2)$. Now, it is easy to find a constant in the last expression, namely, the missing constant is $C = 1/(\sqrt{2\pi}\tilde{\sigma})$.

Note that in the example above, posterior mean $\tilde{\mu}$ is a weighted average of sample average $\bar{x}_n$ and prior mean μ_0 , i.e. $\tilde{\mu} = \omega_1\bar{x}_n + \omega_2\mu_0$ with $\omega_1 + \omega_2 = 1$ where $\omega_1 = (n/\sigma^2)/(n/\sigma^2 + 1/\tau^2)$ and $\omega_2 = (1/\tau^2)/(n/\sigma^2 + 1/\tau^2)$. Here $1/\tau^2$ may be interpreted as the precision of initial information. If initial information is very precise, i.e. $1/\tau^2$ is large (or τ^2 is small), then the prior mean gets almost all weight, and $\tilde{\mu}$ is close to μ_0 . Thus, we almost ignore new information in the form of a sample $X_1, \dots, X_n$. If initial information is poor, i.e. $1/\tau^2$ is small, prior mean gets almost no weight, and $\tilde{\mu}$ is close to $\bar{x}_n$. Note that as $n \rightarrow \infty$, information from the sample dominates prior information and $\tilde{\mu} \rightarrow \bar{x}_n$. Moreover, at least in our example, as $n \rightarrow \infty$, $\tilde{\sigma}^2 \rightarrow 0$. Thus, as the sample size increases, the posterior distribution converges to a degenerate distribution concentrated at the true parameter value.

Example The same trick works outside the normal family. Let $X = (X_1, \dots, X_n)$ be a random sample from a $\text{Poisson}(\lambda)$ distribution, so that the joint pdf of the data is $f(x|\lambda) = \prod_{i=1}^n e^{-\lambda} \lambda^{x_i}/x_i!$ where $x = (x_1, \dots, x_n)$. Let the prior distribution for λ be the $\text{Gamma}(\alpha, \beta)$ distribution with density

$$\pi(\lambda) = \frac{\lambda^{\alpha-1} \exp\{-\lambda/\beta\}}{\Gamma(\alpha)\beta^\alpha} I\{\lambda > 0\},$$

where $\alpha > 0$ and $\beta > 0$; its mean is $\alpha\beta$ and its variance is $\alpha\beta^2$. Keeping track only of multiplicative terms that contain λ ,

$$\begin{aligned}
\pi(\lambda|X=x) &\propto \pi(\lambda)f(x|\lambda) \\
&\propto \lambda^{\alpha-1} \exp\{-\lambda/\beta\} \cdot e^{-n\lambda} \lambda^{\sum_{i=1}^n x_i} \\
&\propto \lambda^{\alpha+\sum_{i=1}^n x_i-1} \exp\left\{-\lambda\left(n + \frac{1}{\beta}\right)\right\}.
\end{aligned}$$

The last expression is the kernel of a Gamma density. Thus, the conditional distribution of λ given $X = x$ is $\text{Gamma}(\tilde{\alpha}, \tilde{\beta})$ with

$$\tilde{\alpha} = \alpha + \sum_{i=1}^n x_i, \quad \tilde{\beta} = \frac{1}{n + 1/\beta} = \frac{\beta}{n\beta + 1}.$$

The posterior mean is

$$\mathbb{E}[\lambda|X = x] = \tilde{\alpha}\tilde{\beta} = \frac{\beta(\alpha + \sum_{i=1}^n x_i)}{n\beta + 1} = \omega_1 \bar{x}_n + \omega_2 \alpha\beta,$$

where $\omega_1 = n\beta/(n\beta + 1)$ and $\omega_2 = 1/(n\beta + 1)$ with $\omega_1 + \omega_2 = 1$. As in the normal example, the posterior mean is a weighted average of the sample average $\bar{x}_n$ and the prior mean $\alpha\beta$, and as $n \rightarrow \infty$ the information from the sample dominates prior information: $\omega_1 \rightarrow 1$ and the posterior mean approaches $\bar{x}_n$. Similarly, the posterior variance is

$$V(\lambda|X = x) = \tilde{\alpha}\tilde{\beta}^2 = \frac{\beta^2(\alpha + n\bar{x}_n)}{(n\beta + 1)^2},$$

which is of order $\bar{x}_n/n$ for large n and hence tends to zero. Thus, the posterior again converges to a degenerate distribution concentrated at the true parameter value as the sample size increases.

Once we have the posterior distribution, we can construct an estimator of the parameter of interest. Common examples include posterior mean and posterior mode (posterior mode denotes the point with the greatest pdf value of the posterior distribution).

An important problem with Bayesian estimation is that if a prior distribution puts zero probability mass on the true parameter value, then no matter how large our sample is, posterior distribution will put zero mass on the true parameter value as well. You will see examples of this phenomenon in 14.384.

2.2 Conjugate Classes

Let $\mathcal{F}$ be the class of distributions indexed by θ . Let $\mathcal{P}$ be the class of prior distributions of θ . Then we say that $\mathcal{P}$ is conjugate to $\mathcal{F}$ if whenever the data is distributed according to $\mathcal{F}$ and prior distribution is from $\mathcal{P}$, then the posterior distribution is from $\mathcal{P}$ as well. For example, we have already seen that the class of normal distributions is conjugate to the class of normal distributions with known (fixed) variance. It is also known that the class of $\mathcal{B}$ -distributions is conjugate to the class of binomial distributions. We have also just seen that the class of Gamma distributions is conjugate to the class of Poisson distributions. The concept of conjugate classes is introduced because of its mathematical convenience. It is relatively easy to calculate the posterior when the prior lies in the conjugate class.

Remark 1. The two conjugate pairs above are not coincidences; they are instances of a general recipe. Recall from Lecture Note 5 that a k -parameter exponential family has density $f(x; \theta) =$

$h(x) c(\theta) \exp \left(\sum_{j=1}^k w_j(\theta) t_j(x) \right)$. For such a family, consider priors of the form

$$\pi(\theta) \propto c(\theta)^a \exp \left(\sum_{j=1}^k w_j(\theta) b_j \right),$$

indexed by hyperparameters $(a, b_1, \dots, b_k)$. Multiplying the prior by the likelihood of an i.i.d. sample $x_1, \dots, x_n$ gives

$$\pi(\theta | x) \propto c(\theta)^{a+n} \exp \left(\sum_{j=1}^k w_j(\theta) \left(b_j + \sum_{i=1}^n t_j(x_i) \right) \right),$$

which is again in the same class, with updated hyperparameters $a \mapsto a + n$ and $b_j \mapsto b_j + \sum_i t_j(x_i)$. Every exponential family therefore has a natural conjugate class, and Bayesian updating within it reduces to updating hyperparameters by sufficient statistics. The Beta prior for the Bernoulli and the Normal prior for the Normal mean are exactly this construction.

2.3 Simulation Techniques

Conjugate priors were almost the only priors used for a long time, as the others tend to be not analytically tractable. Nowadays, there are numerical algorithms (MCMC — Markov Chain Monte Carlo) that allow one to calculate the posterior for priors outside the conjugate family. MCMC will be discussed in 14.384 and 14.385.

The goal of a typical MCMC algorithm is to get a sequence of random draws from the posterior distribution. That is, to construct a numerical algorithm that would produce $\theta_1, \dots, \theta_B$ as simulated from $\pi(\theta | X = x)$ (not always independent draws, but draws that may be stationary and satisfy a Law of Large Numbers). This allows one to make inferences of any sort on θ or any function of θ . Imagine that you wish to use the mean of the posterior as an estimator; then $\frac{1}{B} \sum_{b=1}^B \theta_b$ is an estimator for θ , while $\frac{1}{B} \sum_{b=1}^B g(\theta_b)$ is an estimator for parameter $\tau = g(\theta)$. If you prefer to use the median of the posterior as an estimate, then you would order the draws θ_b in increasing order and take the middle one $\theta_{([B/2])}$ as an estimate for θ , etc.

2.4 Credible Intervals

The posterior distribution contains all the information that a researcher can get from the data. However, it is often impractical to report a posterior distribution, as it might be intractable or it might not have analytic form at all. Therefore, it is a common practice to report only some characteristics of a posterior distribution, such as its mean, variance, and mode. Mean and mode of the posterior serve as point estimators of the parameter while variance shows the precision of the posterior distribution. A better way to show the precision of the posterior distribution is the concept of *credible intervals*. Let X be our data, $\theta \in \Theta$ our parameter, and $\pi(\theta | X = x)$ the posterior distribution. Then for any $\alpha \in [0, 1]$, a set $C(x) \subset \Theta$ is called $1 - \alpha$ -credible if $\pi(\theta \in C(x) | X = x) \geq 1 - \alpha$. In words, the set $C(x)$ contains the true parameter value θ with

probability at least $1 - \alpha$. Of course, the whole parameter space Θ is $1 - \alpha$ -credible. But, clearly, reporting Θ as a $1 - \alpha$ -credible set is not useful at all. The smallest $1 - \alpha$ -credible set contains only points with the highest posterior density.

As an example, let $X = (X_1, \dots, X_n)$ be a random sample from $N(\mu, \sigma^2)$ with σ^2 known and let $N(\mu_0, \tau^2)$ be the prior distribution of μ . We have already seen that the conditional distribution of μ given $X = x$ is $N(\tilde{\mu}, \tilde{\sigma}^2)$ with $\tilde{\mu}$ and $\tilde{\sigma}^2$ defined as above. Then $(\mu - \tilde{\mu})/\tilde{\sigma} \sim N(0, 1)$. Let z_γ be the γ -quantile of the standard normal distribution. Then $\pi\{(\mu - \tilde{\mu})/\tilde{\sigma} \in [z_{\alpha/2}, -z_{\alpha/2}] | X = x\} = 1 - \alpha$ or, equivalently, $\pi\{\tilde{\mu} + z_{\alpha/2}\tilde{\sigma} \leq \mu \leq \tilde{\mu} - z_{\alpha/2}\tilde{\sigma} | X = x\} = 1 - \alpha$. Since the pdf $\phi(x)$ of the standard normal distribution is decreasing on $x \geq 0$ and increasing on $x \leq 0$, $[\tilde{\mu} + z_{\alpha/2}\tilde{\sigma}, \tilde{\mu} - z_{\alpha/2}\tilde{\sigma}]$ is the shortest $1 - \alpha$ -credible interval.

A credible set of level $1 - \alpha$ is not unique: any region holding $1 - \alpha$ of the posterior probability qualifies, so a choice must be made. Two choices are standard. The *equal-tailed* credible interval runs from the $\alpha/2$ to the $1 - \alpha/2$ quantile of the posterior, leaving equal probability in each tail. The *highest posterior density* (HPD) set instead collects the parameter values with the largest posterior density — lower a horizontal cut through the posterior density until the region above the cut holds probability $1 - \alpha$ — and is, by construction, the shortest region with the required probability. When the posterior is symmetric and unimodal, as in the Normal-Normal example, the two coincide. When the posterior is skewed, they differ: the HPD set shifts toward the mode and can be noticeably shorter, while the equal-tailed interval is simpler to compute (two quantiles) and is invariant to monotone reparameterizations of θ , which the HPD set is not.

3 Large Sample Properties of Bayes' Procedures

There is a mathematical theorem which claims that under some regularity conditions the prior vanishes asymptotically and we get essentially MLE inferences.

Theorem 1. *Under appropriate regularity conditions the posterior is approximately normal with mean $\hat{\theta}_{ML}$ and variance $\frac{1}{nI_1(\hat{\theta}_{ML})}$ when n is large. In particular, the asymptotic frequentist $1 - \alpha$ interval*

$$C = \left[\hat{\theta}_{ML} + z_{\alpha/2} \sqrt{\frac{1}{nI_1(\hat{\theta}_{ML})}}, \hat{\theta}_{ML} - z_{\alpha/2} \sqrt{\frac{1}{nI_1(\hat{\theta}_{ML})}} \right]$$

is also an approximate Bayesian credible set at level $1 - \alpha$:

$$P(\theta \in C | X = x) \rightarrow 1 - \alpha \text{ as } n \rightarrow \infty$$

The theorem says that the posterior concentrates around the asymptotic limit of frequentist MLE, and in any “reasonable” situation Bayes estimate in large samples will be close to MLE. However, you should be cautious about this theorem!

Cautions:

- The theorem is about asymptotics. However, the prior can influence inferences in finite samples.

- One of the regularity assumptions is the identification condition. If you are not identified, then where the Bayesian estimator converges depends on your prior.
- The prior should not restrict the parameter space unnecessarily.

The strongest critique and objection people may have to Bayes inference is the question of where priors come from. Priors are subjective by definition, they summarize one's subjective opinion or belief about possible values of θ . There is a general desire for non-informative priors, which is a hard question.

3.1 Bayesian Testing

Let $\pi(\theta|X = x)$ be our posterior distribution. One can test the hypothesis $H_0 : \theta \in \Theta_0$ against the hypothesis that $H_1 : \theta \in \Theta_1$ by looking at the posterior odds and accepting the null iff:

$$P(\theta \in \Theta_0|X = x) \geq P(\theta \in \Theta_1|X = x).$$

It is interesting that the hypotheses are treated in a symmetric way here. You would face a difficulty if you want to test a specific parameter value, for example, $H_0 : \theta = \theta_0$, as the continuous posterior puts zero weight on any point. In such a case one has to put a point mass of prior weight on this specific value for testing.